POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6033D | Semester VI | Theory | 4 Credits

Power System Economics and Distributed Generation

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6033D as Power System Economics and Distributed Generation in Semester VI for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kanpur’s Economic Operations and Control of Power Systems course and polytechnic power system curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Economic models, tariff calculations, grid integration designs and DG installations must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support load curve analysis, economic dispatch, DG technologies, and microgrid concepts. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Power System Economics and Distributed Generation studies the economic and technical aspects of power generation, dispatch and decentralized energy systems. It develops the ability to analyze load curves, evaluate power generation costs and tariffs, understand economic load dispatch, and coordinate the integration of distributed generation (DG) and microgrids while respecting the technical and safety boundaries of electrical and power engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Power systems I & II, electrical machines, basic economics and energy conversion principles.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Analyze load curves and calculate the economic parameters of power generation including cost and tariffs.
2. Apply economic load dispatch and unit commitment principles for optimal power system operation.
3. Explain the technologies, advantages and challenges of distributed generation (DG) systems.
4. Analyze the technical impacts and integration requirements for microgrids and DG in modern power systems.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Analyze load curves and calculate power generation costs and tariff structures.	Apply
CO2	Apply economic load dispatch principles for optimal power system operation.	Apply
CO3	Explain the technologies and grid integration requirements for distributed generation.	Understand
CO4	Explain the concepts of microgrids and energy storage in modern power systems.	Understand

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7-PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Power Generation Economics and Tariffs	Load curve analysis: Daily, monthly and annual load curves; Load factor, Diversity factor, Plant capacity factor; Economics of power generation: Capital and Operating costs; Tariffs: Flat rate, Block rate, Two-part tariff.	Approx. 14 hours	CO1	Module 1
2	Economic Operation of Power Systems	Economic Load Dispatch (ELD): Input-output characteristics, Incremental fuel cost; ELD neglecting and considering transmission losses; Unit Commitment (UC) basics; Optimal Power Flow (OPF) concepts.	Approx. 16 hours	CO2	Module 2
3	Distributed Generation Technologies	Definition and rationale for Distributed Generation (DG); DG technologies: Solar PV, Wind turbines, Fuel cells, Micro-turbines, Biomass; Advantages and environmental impacts of DG.	Approx. 14 hours	CO3	Module 3
4	Grid Integration and Microgrids	Technical impacts of DG: Voltage profile, Protection coordination, Power quality; Microgrid concept: Architecture, Islanded vs Grid-connected modes; Energy storage in microgrids; V2G basics.	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Power Generation Economics and Tariffs

Load curve analysis: Daily, monthly and annual load curves; Load factor, Diversity factor, Plant capacity factor; Economics of power generation: Capital and Operating costs; Tariffs: Flat rate, Block rate, Two-part tariff.

CO1 · Approx. 14 hours

Economic Operation of Power Systems

Economic Load Dispatch (ELD): Input-output characteristics, Incremental fuel cost; ELD neglecting and considering transmission losses; Unit Commitment (UC) basics; Optimal Power Flow (OPF) concepts.

CO2 · Approx. 16 hours

Distributed Generation Technologies

Definition and rationale for Distributed Generation (DG); DG technologies: Solar PV, Wind turbines, Fuel cells, Micro-turbines, Biomass; Advantages and environmental impacts of DG.

CO3 · Approx. 14 hours

Grid Integration and Microgrids

Technical impacts of DG: Voltage profile, Protection coordination, Power quality; Microgrid concept: Architecture, Islanded vs Grid-connected modes; Energy storage in microgrids; V2G basics.

CO4 · Approx. 16 hours

MODULE 1

Power Generation Economics and Tariffs

Load curve analysis: Daily, monthly and annual load curves; Load factor, Diversity factor, Plant capacity factor; Economics of power generation: Capital and Operating costs; Tariffs: Flat rate, Block rate, Two-part tariff.

Approx. 14 hours | CO1 | Understand · Apply

Learning goals

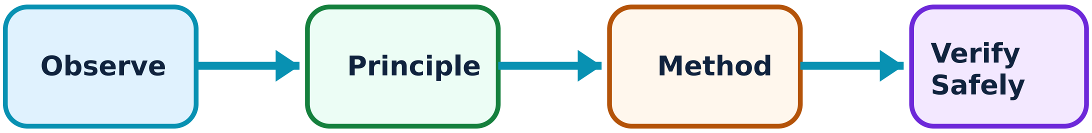
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Power Generation Economics and Tariffs: identify the system, state its principle, follow the correct method and verify the result safely.

1. Load curve and factors−

Load curves show the variation of electrical load over time. Key factors like 'Load Factor' (average load / peak load) and 'Diversity Factor' (sum of individual max demands / simultaneous max demand) help in determining the capacity and efficiency of the power plant.

Study and application method

Calculate the 'Load Factor' and 'Plant Capacity Factor' for a given load profile; explain the significance of a high 'Diversity Factor'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the 'Load Factor' and 'Plant Capacity Factor' for a given load profile; explain the significance of a high 'Diversity Factor'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Load data must be accurate for reliable system planning. Do not use estimated load curves for safety-critical capacity planning without authorised verification.

2. Economics and Tariffs−

The cost of power generation includes fixed capital costs and variable operating costs. Tariffs are the rates at which energy is sold to consumers. A 'Two-part Tariff' consists of a fixed charge based on maximum demand and a variable charge based on actual energy consumption.

Study and application method

Differentiate between 'Fixed Costs' and 'Running Costs' of a power plant; explain the structure of a 'Block Rate Tariff'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Fixed Costs' and 'Running Costs' of a power plant; explain the structure of a 'Block Rate Tariff'.

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Common mistake / precaution: Tariff structures are regulated by utility commissions. Do not provide billing or financial advice without authorised review of the latest official utility rate schedules.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Economic Operation of Power Systems

Economic Load Dispatch (ELD): Input-output characteristics, Incremental fuel cost; ELD neglecting and considering transmission losses; Unit Commitment (UC) basics; Optimal Power Flow (OPF) concepts.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

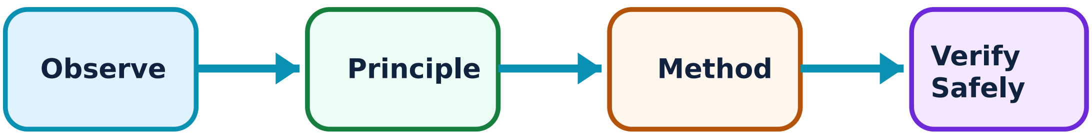
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Economic Operation of Power Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Economic Load Dispatch (ELD)−

ELD aims to distribute the total load among available generating units to minimize the total fuel cost. This is achieved by operating units at equal 'Incremental Fuel Costs'. When transmission losses are considered, 'Penalty Factors' are used to adjust the dispatch.

Study and application method

State the 'Equal Incremental Cost' criterion for ELD; explain how 'Transmission Losses' affect the optimal power distribution.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State the 'Equal Incremental Cost' criterion for ELD; explain how 'Transmission Losses' affect the optimal power distribution.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: ELD calculations assume stable system operation. Do not implement dispatch changes without authorised verification of system frequency and voltage stability.

2. Unit Commitment and Optimization−

Unit Commitment (UC) involves deciding which generating units should be ON or OFF over a period to meet the load at minimum cost while satisfying constraints like minimum up/down time. OPF integrates ELD with power flow constraints to ensure secure operation.

Study and application method

Describe the 'Unit Commitment' problem and its constraints; explain the difference between 'ELD' and 'Optimal Power Flow' (OPF).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the 'Unit Commitment' problem and its constraints; explain the difference between 'ELD' and 'Optimal Power Flow' (OPF).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Optimization models are simplified representations. All operational decisions must follow authorised grid codes and real-time security assessments.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Distributed Generation Technologies

Definition and rationale for Distributed Generation (DG); DG technologies: Solar PV, Wind turbines, Fuel cells, Micro-turbines, Biomass; Advantages and environmental impacts of DG.

Approx. 14 hours **CO3** **Understand · Apply**

Learning goals

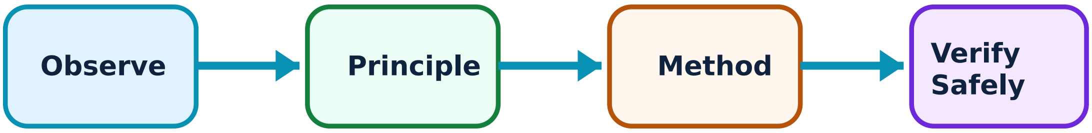
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Distributed Generation Technologies: identify the system, state its principle, follow the correct method and verify the result safely.

1. Introduction to DG−

Distributed Generation refers to small-scale power generation located close to the point of consumption. It reduces transmission losses, improves reliability, and allows for the integration of renewable energy. DG units can be grid-connected or stand-alone.

Study and application method

List four advantages of 'Distributed Generation' over centralized power systems; identify three common 'DG Technologies'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List four advantages of 'Distributed Generation' over centralized power systems; identify three common 'DG Technologies'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: DG units must be properly synchronized with the grid. Do not connect DG systems without authorised grid-interface protection and utility approval.

2. DG Technologies and Impact−

Solar PV and Wind are intermittent sources requiring energy storage. Fuel cells and micro-turbines provide more stable output. DG helps in reducing carbon emissions but poses challenges in voltage regulation and protection coordination.

Study and application method

Explain the role of 'Fuel Cells' in distributed generation; describe the environmental benefits of using 'Biomass' for local power generation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the role of 'Fuel Cells' in distributed generation; describe the environmental benefits of using 'Biomass' for local power generation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper DG integration can cause voltage swells or protection failure. All DG installations must follow authorised technical standards for grid interconnection.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Grid Integration and Microgrids

Technical impacts of DG: Voltage profile, Protection coordination, Power quality; Microgrid concept: Architecture, Islanded vs Grid-connected modes; Energy storage in microgrids; V2G basics.

Approx. 16 hours | CO4 | Understand · Apply

Learning goals

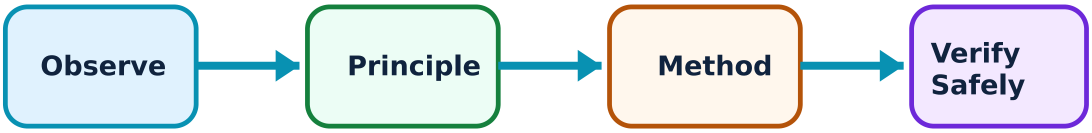
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Grid Integration and Microgrids: identify the system, state its principle, follow the correct method and verify the result safely.

1. Microgrid Architecture and Operation−

A Microgrid is a localized group of electricity sources and loads that can operate connected to the main grid or independently (Islanded mode). It includes DG units, energy storage (batteries), and a control system to manage power flow and stability.

Study and application method

Differentiate between 'Islanded Mode' and 'Grid-connected Mode' of a microgrid; sketch a basic 'Microgrid Architecture'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Islanded Mode' and 'Grid-connected Mode' of a microgrid; sketch a basic 'Microgrid Architecture'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Transitioning between modes requires precise control. Do not operate islanded microgrids without authorised 'Black Start' capabilities and stable frequency/voltage control.

2. Integration Challenges and Storage−

Integrating DG requires managing power quality issues like harmonics and flicker. Energy storage systems (ESS) are critical for balancing supply and demand in microgrids. 'Vehicle-to-Grid' (V2G) allows EVs to act as distributed storage for the grid.

Study and application method

Explain the importance of 'Energy Storage' in a microgrid with renewable sources; describe the 'Islanding' phenomenon and its safety implications.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the importance of 'Energy Storage' in a microgrid with renewable sources; describe the 'Islanding' phenomenon and its safety implications.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Unintentional islanding is a major safety hazard for utility workers. All DG systems must include authorised 'Anti-islanding' protection verified by the utility.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Power Generation Economics and Tariffs

Use the concept flow to revise observation, principle, method and verification.

Module 2: Economic Operation of Power Systems

Use the concept flow to revise observation, principle, method and verification.

Module 3: Distributed Generation Technologies

Use the concept flow to revise observation, principle, method and verification.

Module 4: Grid Integration and Microgrids

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Centralized and Distributed generation in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the daily load curve for a typical residential area and label the base and peak loads.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the block diagram of a Microgrid showing DG units, loads, and the Point of Common Coupling (PCC).

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the incremental fuel cost conceptually for a generator with a given cost function.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a local distributed generation project (e.g., rooftop solar) and note its capacity and grid connection type.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Power Generation Economics and Tariffs

Explain Load curve and factors and state one correct method, application or precaution.−

Load curves show the variation of electrical load over time. Key factors like 'Load Factor' (average load / peak load) and 'Diversity Factor' (sum of individual max demands / simultaneous max demand) help in determining the capacity and efficiency of the power plant.

Answer framework: Calculate the 'Load Factor' and 'Plant Capacity Factor' for a given load profile; explain the significance of a high 'Diversity Factor'.

Important point: Load data must be accurate for reliable system planning. Do not use estimated load curves for safety-critical capacity planning without authorised verification.

Explain Economics and Tariffs and state one correct method, application or precaution.−

The cost of power generation includes fixed capital costs and variable operating costs. Tariffs are the rates at which energy is sold to consumers. A 'Two-part Tariff' consists of a fixed charge based on maximum demand and a variable charge based on actual energy consumption.

Answer framework: Differentiate between 'Fixed Costs' and 'Running Costs' of a power plant; explain the structure of a 'Block Rate Tariff'.

Important point: Tariff structures are regulated by utility commissions. Do not provide billing or financial advice without authorised review of the latest official utility rate schedules.

Module 2 — Economic Operation of Power Systems

Explain Economic Load Dispatch (ELD) and state one correct method, application or precaution.−

ELD aims to distribute the total load among available generating units to minimize the total fuel cost. This is achieved by operating units at equal 'Incremental Fuel Costs'. When transmission losses are considered, 'Penalty Factors' are used to adjust the dispatch.

Answer framework: State the 'Equal Incremental Cost' criterion for ELD; explain how 'Transmission Losses' affect the optimal power distribution.

Important point: ELD calculations assume stable system operation. Do not implement dispatch changes without authorised verification of system frequency and voltage stability.

Explain Unit Commitment and Optimization and state one correct method, application or precaution.−

Unit Commitment (UC) involves deciding which generating units should be ON or OFF over a period to meet the load at minimum cost while satisfying constraints like minimum up/down time. OPF integrates ELD with power flow constraints to ensure secure operation.

Answer framework: Describe the 'Unit Commitment' problem and its constraints; explain the difference between 'ELD' and 'Optimal Power Flow' (OPF).

Important point: Optimization models are simplified representations. All operational decisions must follow authorised grid codes and real-time security assessments.

Module 3 — Distributed Generation Technologies

Explain Introduction to DG and state one correct method, application or precaution.−

Distributed Generation refers to small-scale power generation located close to the point of consumption. It reduces transmission losses, improves reliability, and allows for the integration of renewable energy. DG units can be grid-connected or stand-alone.

Answer framework: List four advantages of 'Distributed Generation' over centralized power systems; identify three common 'DG Technologies'.

Important point: DG units must be properly synchronized with the grid. Do not connect DG systems without authorised grid-interface protection and utility approval.

Explain DG Technologies and Impact and state one correct method, application or precaution.−

Solar PV and Wind are intermittent sources requiring energy storage. Fuel cells and micro-turbines provide more stable output. DG helps in reducing carbon emissions but poses challenges in voltage regulation and protection coordination.

Answer framework: Explain the role of 'Fuel Cells' in distributed generation; describe the environmental benefits of using 'Biomass' for local power generation.

Important point: Improper DG integration can cause voltage swells or protection failure. All DG installations must follow authorised technical standards for grid interconnection.

Module 4 — Grid Integration and Microgrids

Explain Microgrid Architecture and Operation and state one correct method, application or precaution.−

A Microgrid is a localized group of electricity sources and loads that can operate connected to the main grid or independently (Islanded mode). It includes DG units, energy storage (batteries), and a control system to manage power flow and stability.

Answer framework: Differentiate between 'Islanded Mode' and 'Grid-connected Mode' of a microgrid; sketch a basic 'Microgrid Architecture'.

Important point: Transitioning between modes requires precise control. Do not operate islanded microgrids without authorised 'Black Start' capabilities and stable frequency/voltage control.

Explain Integration Challenges and Storage and state one correct method, application or precaution.−

Integrating DG requires managing power quality issues like harmonics and flicker. Energy storage systems (ESS) are critical for balancing supply and demand in microgrids. 'Vehicle-to-Grid' (V2G) allows EVs to act as distributed storage for the grid.

Answer framework: Explain the importance of 'Energy Storage' in a microgrid with renewable sources; describe the 'Islanding' phenomenon and its safety implications.

Important point: Unintentional islanding is a major safety hazard for utility workers. All DG systems must include authorised 'Anti-islanding' protection verified by the utility.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Power Generation Economics and Tariffs.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Economic Operation of Power Systems.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Distributed Generation Technologies.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Grid Integration and Microgrids.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Load curve analysis: Daily, monthly and annual load curves; Load factor, Diversity factor, Plant capacity factor; Economics of power generation: Capital and Operating costs; Tariffs: Flat rate, Block rate, Two-part tariff..
2. Develop a complete answer or practical plan for: Economic Load Dispatch (ELD): Input-output characteristics, Incremental fuel cost; ELD neglecting and considering transmission losses; Unit Commitment (UC) basics; Optimal Power Flow (OPF) concepts..
3. Develop a complete answer or practical plan for: Definition and rationale for Distributed Generation (DG); DG technologies: Solar PV, Wind turbines, Fuel cells, Micro-turbines, Biomass; Advantages and environmental impacts of DG..
4. Develop a complete answer or practical plan for: Technical impacts of DG: Voltage profile, Protection coordination, Power quality; Microgrid concept: Architecture, Islanded vs Grid-connected modes; Energy storage in microgrids; V2G basics..

LAST-MILE REVISION

Quick Revision

Module 1: Power Generation Economics and Tariffs

Load curve analysis: Daily, monthly and annual load curves; Load factor, Diversity factor, Plant capacity factor; Economics of power generation: Capital and Operating costs; Tariffs: Flat rate, Block rate, Two-part tariff.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Economic Operation of Power Systems

Economic Load Dispatch (ELD): Input-output characteristics, Incremental fuel cost; ELD neglecting and considering transmission losses; Unit Commitment (UC) basics; Optimal Power Flow (OPF) concepts.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Distributed Generation Technologies

Definition and rationale for Distributed Generation (DG); DG technologies: Solar PV, Wind turbines, Fuel cells, Micro-turbines, Biomass; Advantages and environmental impacts of DG.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Grid Integration and Microgrids

Technical impacts of DG: Voltage profile, Protection coordination, Power quality; Microgrid concept: Architecture, Islanded vs Grid-connected modes; Energy storage in microgrids; V2G basics.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Load curve and factors	Load curves show the variation of electrical load over time.
Economics and Tariffs	The cost of power generation includes fixed capital costs and variable operating costs.
Economic Load Dispatch (ELD)	ELD aims to distribute the total load among available generating units to minimize the total fuel cost.
Unit Commitment and Optimization	Unit Commitment (UC) involves deciding which generating units should be ON or OFF over a period to meet the load at minimum cost while satisfying constraints like minimum up/down time.
Introduction to DG	Distributed Generation refers to small-scale power generation located close to the point of consumption.
DG Technologies and Impact	Solar PV and Wind are intermittent sources requiring energy storage.
Microgrid Architecture and Operation	A Microgrid is a localized group of electricity sources and loads that can operate connected to the main grid or independently (Islanded mode).
Integration Challenges and Storage	Integrating DG requires managing power quality issues like harmonics and flicker.

LEARNING RESOURCES

References

Recommended power system economics references

1. Abhijit Chakrabarti, Power System Analysis: Operation and Control.
2. D. P. Kothari and I. J. Nagrath, Modern Power System Analysis.
3. H. Lee Willis, Distributed Power Generation: Planning and Evaluation.
4. S. Chowdhury and P. Crossley, Microgrids and Active Distribution Networks.

Approved public power system economics sources

- https://nptel.ac.in/courses/108104191
- https://onlinecourses.nptel.ac.in/noc23_ee128/preview

These sources provide the verified study basis while official Course 6033D detail is unavailable; they do not replace official utility pricing, authorised grid management plans, certified equipment specifications or authorised energy market regulations.